

UNIT-4

UNDERSTANDING AGILE

Agile methodology is a project management framework that breaks projects down into several dynamic phases, commonly known as sprints. In this article, get a high-level overview of Agile project management, plus a few common frameworks to choose the right one for your team.

Scrum, Kanban, waterfall, Agile.

There's a wide array of project management frameworks you can use. Some methods though, like waterfall, aren't as effective for software teams. With priorities and customer needs constantly changing, the Agile methodology breaks projects up into several phases to drive continuous improvement.

Agile project management isn't just useful for software development—all types of teams have been successful with this dynamic methodology. If you're looking to get started with Agile methodology, you've come to the right place.

Agile methodology is a project management framework that breaks projects down into several dynamic phases, commonly known as sprints.

The Agile framework is an iterative methodology. After every sprint, teams reflect and look back to see if there was anything that could be improved so they can adjust their strategy for the next sprint.

What are the 4 pillars of Agile?

As outlined in the **Agile Manifesto**, there are **four** main **values** of Agile project management:

1. Individuals over processes and tools. Agile teams value team collaboration and teamwork over working independently and doing things "by the book."
2. Working software over comprehensive documentation. The software that Agile teams develop should work. Additional work, like documentation, is not as important as developing good software.
3. Customer collaboration over contract negotiation. Customers are extremely important within the Agile methodology. Agile teams allow customers to guide where the software should go. Therefore, customer collaboration is more important than the finer details of contract negotiation.

4. Responding to change over following a plan. One of the major benefits of Agile project management is that it allows teams to be flexible. This framework allows for teams to quickly shift strategies and workflows without derailing an entire project.

What are the 12 Agile principles?

If the four values of the Agile model are the weight-bearing pillars of a house, then the 12 Agile principles are the rooms you can build within that house. These principles can be easily adapted to fit the needs of your software development process.

The 12 principles used in Agile methodology are:

1. Satisfy customers through early, continuous improvement and delivery. When customers receive new updates regularly, they're more likely to see the changes they want within the product. This leads to happier, more satisfied customers—and more recurring revenue.
2. Welcome changing requirements, even late in the project. The Agile framework is all about adaptability. In iterative approaches like Agile, being inflexible causes more harm than good.
3. Deliver value frequently. Similar to principle #1, continuous delivery of value to your customers or stakeholders frequently makes it less likely for them to churn.
4. Break the silos of your projects. Cross-functional teams and collaboration is a key Agile value. The goal is for people to break out of their individual projects and collaborate more frequently.
5. Build projects around motivated individuals. Agile management works best when teams are committed and actively working to achieve a goal.
6. The most effective way to communicate is face-to-face. If you're working on a distributed team, spend time communicating in ways that involve face-to-face communication like Zoom calls or daily stand-up meetings.
7. Working software is the primary measure of progress. The ultimate goal of software development projects is a working product, and the Agile framework supports this by prioritizing functional software above all.
8. Maintain a sustainable working pace. Some aspects of Agile project management can be fast-paced, but it shouldn't be so fast that team members burn out. The goal is to maintain sustainability throughout the development process.

9. Continuous excellence enhances agility. If the team develops excellent code in one sprint, they can continue to build off of it the next. Continually creating great work allows teams to move faster in the future.
10. Simplicity is essential. Sometimes the simplest solution is the best solution. Agile development aims to not overcomplicate things and find simple answers to complex problems.
11. Self-organizing teams generate the most value. Similar to principle #5, proactive teams become valuable assets to the company as they strive to deliver continuous improvement.
12. Regularly reflect and adjust your way of work to improve effectiveness. Retrospective meetings are a common Agile practice. It's a dedicated time for Agile teams to look back and reflect on their performance and adapt their behaviors for the

What are the benefits of the Agile development methodology?

You commonly find Agile project management used in application development or other types of software development. This is because software is constantly changing, and the needs of the product have to change with it.

Because of this, linear project management methods like the waterfall model are less effective. Here are a few other reasons why teams use Agile:

Agile methods are adaptable

There's a reason why they call it the Agile methodology. One of the main benefits of using Agile processes in software development is the ability to shift strategies quickly, without disrupting the flow of a project.

Because phases in the traditional waterfall method flow into one another, shifting strategies is challenging and can disrupt the rest of the project roadmap. Since software development is a much more adaptable field, project managing rapid changes in the traditional sense can be challenging. This is part of the reason why Agile project management is favored in software development.

Agile fosters collaborative teamwork

One of the Agile principles states that the most effective way to communicate with your team is face-to-face. Combine this with the principle that encourages teams to break project silos and you have a recipe for collaborative teamwork.

While technology has changed since Agile's inception and work has shifted to welcome more remote-friendly policies, the idea of working face-to-face still hasn't changed.

Agile methods focus on customer needs

One of the unique aspects of software development is that teams can focus on customer needs much more closely than other industries. With the rise of cloud-based software, teams can get feedback from their actual customers quickly. Since customer satisfaction is a key driver for software development, it's easy to see why it was included in the Agile process. By collaborating with customers, Agile teams can prioritize features that focus on customer needs. When those needs change, teams can take an Agile approach and shift to a different project.

Types of Agile methodologies

The Agile framework is an umbrella for several different variations. Here are a few of the most common Agile methodologies.

Kanban

Kanban is a visual approach to Agile. Teams use online Kanban board tools to represent where certain tasks are in the development process. Tasks are represented by cards on a board, and stages are represented in columns. As team members work on tasks, they move cards from the backlog column to the column that represents the stage the task is in.

This method is a good way for teams to identify roadblocks and to visualize the amount of work that's getting done.

Scrum

Scrum is a common Agile methodology for small teams and also involves sprints. The team is led by a Scrum master whose main job is to clear all obstacles for others executing the day-to-day work.

Scrum teams meet daily to discuss active tasks, roadblocks, and anything else that may affect the development team.

- Sprint planning: This event kicks off the sprint. Sprint planning outlines what can be delivered in a sprint (and how).
- Sprint retrospective: This recurring meeting acts as a sprint review—to iterate on learnings from a previous sprint that will improve and streamline the next one.

Extreme Programming (XP)

Typically used in software development, Extreme Programming (XP) is an Agile framework that outlines values that will allow your team to work together more effectively.

The five values of XP include:

- Communication
- Simplicity
- Feedback
- Courage
- Respect

Similar to daily Scrum standups, there are regular releases and iterations, yet XP is much more technical in its approach. If your dev team needs to quickly release and respond to customer requests, XP focuses on the “how” it will get done.

It continually cycles via the entire software development procedure to incrementally, fast and visibly develop an applicable portion of the software. A pivotal component of agile development is rapid subsequent iterations via the software development process. Each iteration via prerequisites, execution, unit testing, and user testing ushers to a purified working product in the following iteration. These methods are engaging because they are more adaptable when product specifications are not set and the span to market is crucial.

Marketing departments’ most standard agile practices comprise user stories, Daily standups, retrospectives, and frequent releases.

However, marketers testing with agile working methods have backers across the association. Many say that the IT/software development operations within their associations also utilize factors of agile methodology. For the latest agile marketing teams, guidance from undergone agile practitioners, like software developers, can be very profitable in the extended term.

No one apparently knows what the future holds, but one thing that is crystal clear is that in the coming years, Non-Agile Marketing departments will plan to implement Agility for benefits.

WHY DO COMPANIES USE AGILE?

The few advantages that it offers are:

- Concentrate on persistent delivery, thereby resulting in more rapid software discharges
- Brisk response to persistent evolutions in the market while the software development procedure
- Enhanced visibility and being capable of pinpointing probable show-stoppers before they become a problem
- Lowering of haggard work / wasted time resulting in augmented productivity and efficiency

EXAMPLES:

1. SKY: It carries a famous name for households like satellite TV, broadband and telephone service. But you know what? The media provider holds a title for them too in software development. The latest and old projects were transformed to Agile, established on the view that software is developed via cooperation between cross-functional and self-organizing teams. It also promotes software to be cast as fast and regularly as achievable. Altering to Agile wasn't manageable, and it interested consequential transformations in attitudes and perspectives and the deduction of well-known procedures.
- PHILIPS: Philips is another company that also follows embraced Agile principles. After considerable modifications to managing structure, the firm presented several Agile coaches that deployed Scrum principles such as Scrum boards and splitting teams into shorter ones. As an outcome of changes like this, teams could also react to situations rapidly and it was ultimately easier for these smaller teams to take duty for their individual creations.

Agile is considered to be the most effective model of contemporary project management because of its flexibility, implementable cooperative approach, and client-oriented approach. AGILE is a methodology that focuses on the flexibility of change and iterative construction; hence, it prepares teams for any changing requirements and conditions. Through fostering timely and regular feedback as well as enhancing cross-team communication, Agile guarantees products' development based on users' needs and demands. This characteristic not only improves efficiency and productivity but also creativity, which is why Agile is widely used for rapidly developing industries.

What is the Agile Model?

The Agile model is a form of project management developed especially for software development that is more flexible in regards to the linear route map and much more focused on customer feedback. As for the advantages of Agile, it has to be mentioned that the process is divided into small segments known as sprints or iterations, where one succeeds the other to provide gradual improvements where none can be made or the requirements are constantly shifting. This approach facilitates closer cooperation between the stakeholders and developers and results in a product that is closer to consumers' needs and market requirements, as well as higher effectiveness and adaptability.

Explain 10 reasons why Agile is the best model

1. **Flexibility to Change:** Scatterable helps new information to be incorporated into the process without affecting the whole project, hence making it flexible to the change of requirements.
2. **Customer-centric:** Agile highly values the client input and engagement system so that the development process closely reflects the consumers' demands and requirements.
3. **Incremental Delivery:** Agile allows accomplishing the work in iterative cycles so that feedback can be received and changes can be incorporated more often, resulting in more effective and relevant results.
4. **Improved Collaboration:** Agile mandates constant communication between and within various functional teams, hence enhancing the integration of an organization's functional teams and operations.
5. **Enhanced Risk Management:** One is that through constant evaluations and the establishment of an agile model of development, which entails

- the constant addressing of problems that may arise until the project is finished, the possibility of project failure is minimized.
6. **Higher Quality Products:** Another advantage performed in Agile is that testing is performed continuously and in parallel with feedback loops, and therefore fewer defects are delivered to the customer.
 7. **Faster Time-to-Market:** As Agile works iteratively, customer value components are worked on at a much faster pace, which helps the product quickly enter the market and gain a competitive edge.
 8. **Greater Transparency:** Agile ensures openness through feedback and reviews, whereby the stakeholders are frequently involved in the project.
 9. **Empowered Teams:** Due to decision-making and self-organization, teams are more motivated in an agile context, and the latter has strong performance.
 10. **Focus on Value:** Agile puts a lot of importance on releasing first the highest-value items; this gives assurance that significant components of the project are delivered efficiently.

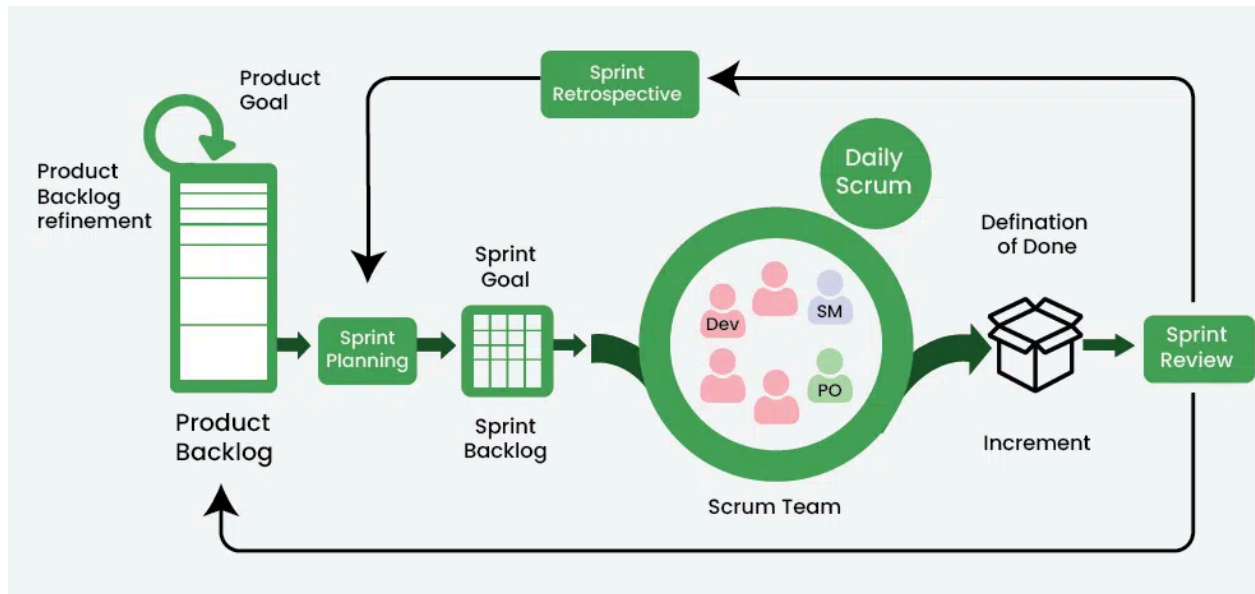
In the ever-evolving world of project management, one term that frequently pops up is "scrum." But what is Scrum? In simple terms, Scrum is a popular framework used in agile project management, designed to enhance collaboration, adaptability, and efficiency within teams.

The Scrum framework has been used since the early 1990s. Scrum is a framework that can be used for building products. This article focuses on discussing Scrum in detail. So, let's delve deeper into what Scrum entails, its key principles, and how it revolutionizes the way teams work together to deliver high-quality results.

What is Scrum?

If we define Scrum in simple terms, then Scrum is a way of managing projects, especially in software development. It's like a playbook that teams use to work together more effectively. Instead of doing everything at once, Scrum breaks work into smaller parts called "sprints." Each sprint focuses on completing a specific piece of the project, allowing teams to adapt and improve as they go. It's all about teamwork, communication, and getting things done step by step.

Scrum is a popular framework that is used whenever we want to develop complex products, **Ken Schwaber** and **Jeff Sutherland** are known as the people who developed Scrum.



- Scrum allows us to develop products of the highest value while making sure that we maintain creativity and productivity.
- The iterative and incremental approach used in scrum allows the teams to adapt to the changing requirements.

The Scrum Framework

The Scrum framework is a simple, agile way to manage projects that helps teams work together to create high-quality products quickly. Here are the basic components and principles of Scrum, explained in simple terms.

Scrum Team

- **Product Owner:** Decides what the team should work on and makes sure it meets business and customer needs.
- **Scrum Master:** Helps the team follow Scrum rules and removes any obstacles they face.
- **Development Team:** A group of people with different skills who work together to build the product.

Artifacts:

- **Product Backlog:** A list of everything that needs to be done for the product, organized by priority.
 - **Sprint Backlog:** A list of tasks the team plans to complete during the current sprint.
 - **Increment:** The finished work at the end of a sprint that could be released to users.
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- Burndown chart is a major parameter used in agile software development and scrum to detect how much work remains to be completed. It is the graphical representation of showing the left-out portion of the task versus time. Generally, time is taken on the abscissa and left out work on ordinates. It is highly used when a project is going to be done. The company gets the knowledge of 'How the team members are working', and "Can determine the accomplishment of the task".

From the burn-down chart graph, we can estimate when the project is going to be complete. It is generally useful when a team does not make any progress in the project in the middle may be due to any reason, at that time they take the help of it and plan accordingly to complete the project within the given time. It checks the productivity of the work. It is widely used in agile and scrum project management.

Events/ Scrum Ceremonies:

Sprint: A short, fixed period (usually 1-4 weeks) where the team works on specific tasks from the sprint backlog.

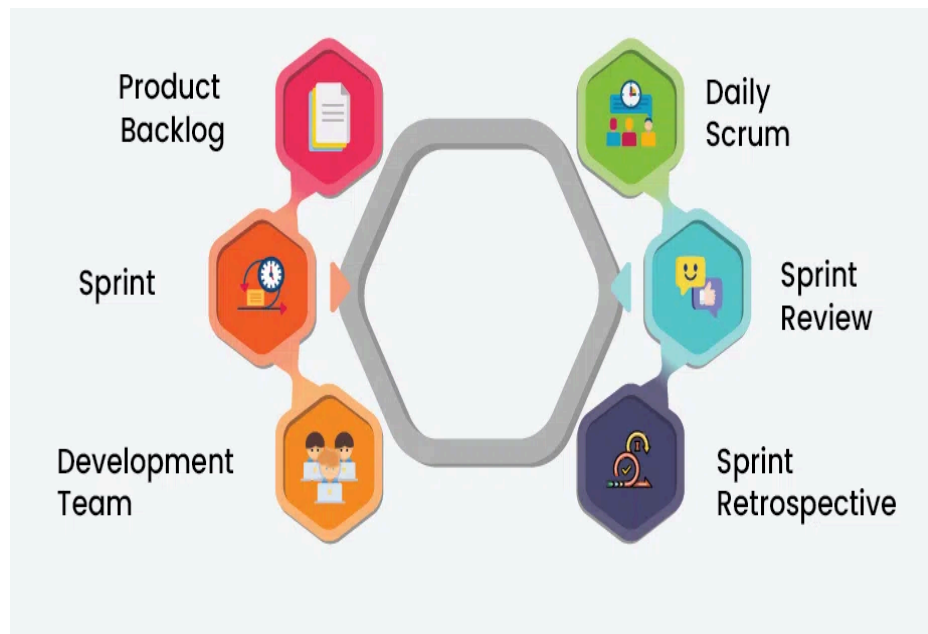
- **Sprint Planning:** A meeting at the start of the sprint where the team decides what to work on.
- **Daily Scrum:** A quick daily meeting where the team checks in on progress and plans the day's work.
- **Sprint Review:** A meeting at the end of the sprint where the team shows what they've accomplished and gets feedback.

- **Sprint Retrospective:** A meeting at the end of the sprint where the team discusses what went well and what could be improved for the next sprint.

Key Terminologies of Scrum

Here are some of key terminologies of Scrum:

1. **Product Backlog:** The product backlog is known to be the prioritized list of fixes as well as features that is included in the product's roadmap.
2. **Sprint:** Sprint is known as a **time-box** event which typically lasts from one week to four weeks, in this phase a product increment or iteration occurs.
3. **Development Team:** The development team is a group of individuals who are professional in their field and are responsible for product delivery.
4. **Daily Scrum:** Daily scrum is a 15 minute daily meeting used by the development team to integrate activities and to create a plan for the next 24 hours of development.
5. **Sprint Review:** The sprint review is held at the end of the sprint in which the team presents all the work that is completed to their stakeholders and the stakeholders give back their feedback.
6. **Sprint Retrospective:** The sprint retrospective is a meeting concluded at the end of each sprint so that the team can discuss what went well and what could be improved as well as how to make those improvements.



What is Scrum Methodology?

Here are some of the following principles and values that characterize Scrum methodology:

Scrum Principles for Project Success

- **Transparency:** When people operate in a team setting, they are aware of the difficulties that other people may be facing. Frequent in-person meetings between project owners and members of cross-functional teams help to avoid misunderstandings and information.
- **Reflection:** The framework includes regular reflection points so that team members can assess their development. Project managers estimate and prepare for the future using the information take from these review meetings. Projects can therefore operate more effectively, on time, and within budget.
- **Adaptation:** Tasks can be rearranged in order by team members in response to evolving client needs. They choose which assignments to finish right away and which to come back to later.

Scrum Values for Project Teams

- **Commitment:** Members of the Scrum Team are devoted to finding the best solution through continuous improvement and time-based tasks and goals.
- **Courage:** By posing direct, difficult questions, Scrum Teams demonstrate bravery. To find the best answer, they have frank and open discussions.
- **Respect:** The Scrum process, the project managers, and each other are respected by the team members. Within the team, this respectful culture fosters a sense of cooperation and mutual support.

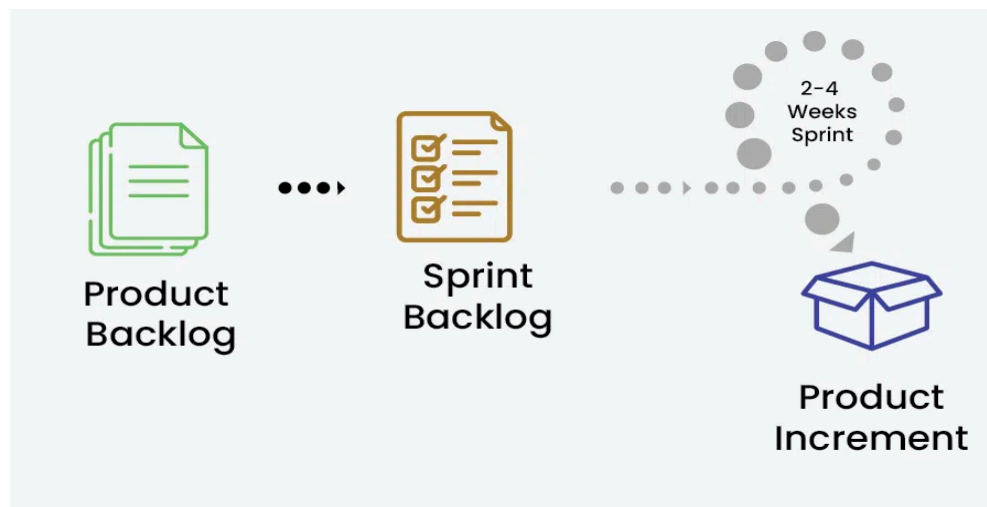
How does Scrum Work?

Scrum divides projects into smaller, more manageable units called "sprints," which span two to four weeks on average. During these sprints, cross-functional teams work together to generate product increments that might potentially be shipped. It uses tools like the product backlog and sprint backlog, as well as regular events like sprint planning, daily stand-ups, sprint reviews, and retrospectives, to promote transparency, inspection, and adaptation. In the end, this helps teams respond rapidly to changes and produce value iteratively.

What are Scrum Artifacts?

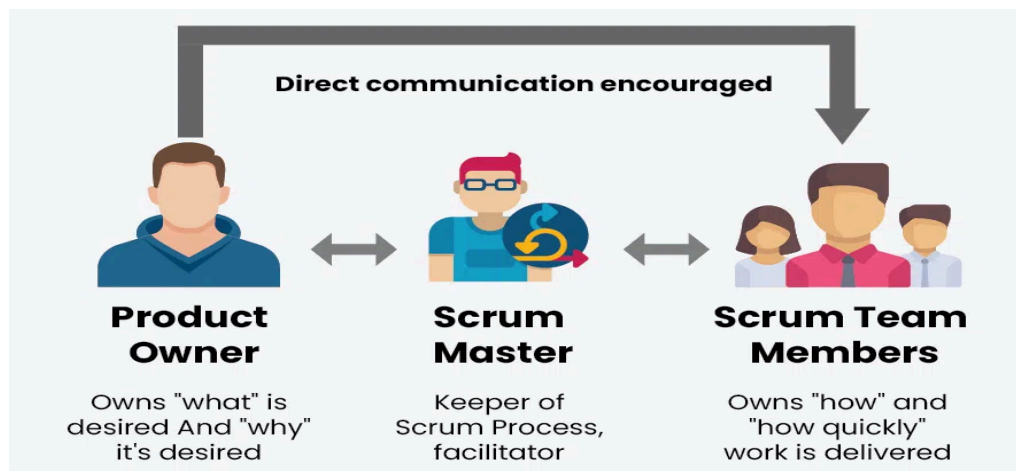
There are three main agile scrum artifacts which are mentioned below:

1. **Sprint Backlog:** The list of everything which the team commits to achieve in a sprint is known as Sprint Backlog. If it is developed once nobody can add to the sprint backlog except the development team.
2. **Product Backlog:** The ordered list of every single thing which is needed in the product based on the product goal is known as product backlog. It is never complete, and it is always evolving.
3. **Product Increments:** Lastly at the end of each sprint the development team delivers a product increment which is potentially releasable.



What are Scrum Roles?

Following are the various roles in the scrum, these are the most commonly known and used widely in the industry:



1. **Scrum Team:** Scrum team is a collection of individuals from the company (typically 6-10 people) who make sure to deliver the project requirements or increments.
2. **Product Owner:** The project owner represents the stakeholders and they are the ones responsible for prioritizing the backlog which ultimately leads to maximum value and team work.
3. **Scrum Master:** The scrum master is an important individual who serves as a guide and mentor to make sure their team understands the scrum framework and its values. The scrum master is highly involved dealing with day to day tasks with their team members.

What are Sprints in Scrum?

Whenever we build projects or products we do it in series of repetitions and these repetitions are known as **sprint**, sprint helps the developers of the project to build projects with higher productivity and functional. Sprints are just a series of steps involved in a time-frame or duration that deals with the overall development of the product.

- The sprint also allows us to adapt to the changes more faster and quickly.
- The sprint helps creators and developers to build products faster.
- The sprint is a series or cycle to break down the tasks.

What is Sprint Process?

Step 1: The scrum cycle is a complete cycle of product development which starts with the **Product Owner**, the product owner defines all the product backlog and requirements for further development.

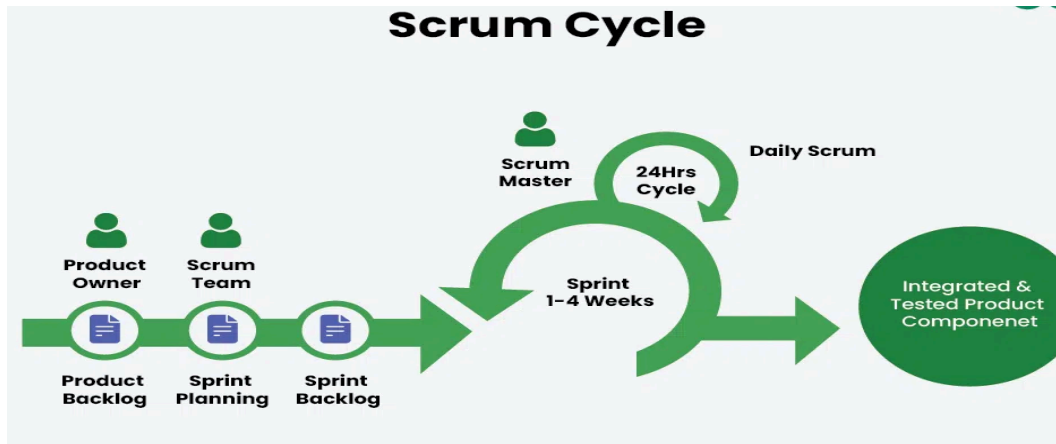
Step 2: The next step is handled by the scrum team and the scrum team manages the **Sprint Planning** in the sprint planning phase the team plans all the tasks and roles for the development of the product.

Step 3: Then in the next phase the team deals with the sprint backlogs which are due and deals accordingly with all the backlogs left.

Step 4: As we know the **scrum master** is someone who oversees all the tasks of the team and acts as a mentor for the team, so in this step the scrum master oversees and collaborates with the team to make sure the product is ready on time, this cycle typically lasts from 1 to 4 weeks but depending on the project size it varies a lot, which is why sometimes for larger projects, it's also divided into smaller part known as **Daily Scrum**.

Step 5: This is the final process stage, in this stage the team integrates all the modules of the project and tests the various components of the product.

Below is the visual representation of the Scrum's sprint process:



Applications of Scrum

We use the scrum framework because it provides us with a flexibility and an iterative approach to do product development and whenever we do product or software development.

The scrum framework allows us to respond with the teams quickly when we have changing requirements which helps us in delivering values incrementally let's understand some common applications of scrum:

1. Software Development

- The scrum framework is popularly used in the [software development field](#) in order to manage and lever deliver software projects of small sizes as well as large and complex projects.
- We use scrum framework because it allows us to collaborate with the development team in order to work with short iterations as we have discussed before these short iterations are called sprints which lasts from two to four weeks at the end of each sprint we get a potential product.

2. Scrum in Marketing

The scrum framework can also be used in the marketing in order to manage and execute various campaigns we can use the scrum because it enables marketing professionals so that they can respond quickly to the changes of the market and

according to those changes they can test different strategies and then optimise their campaigns more efficiently.

3. Social & Non Profit Sector

This framework can also be applied in the social sector as well as the non profit organization's industry to manage various projects efficiently as it allows these organisations to be more adaptive and to be more responsive to the various needs of the community that they serve which can lead to overall growth.

4. Any Complex Project

No matter what sector we are working on the scrum is suitable for any project that deals with complex requirements which are changing with time because when we are working with the product that has changing requirements then using an incremental approach is always beneficial.

5. R&D (Research and Development Sector)

The scrum framework is useful in under industries where we have the work of research and development as we can use the scrum framework to manage ongoing projects and prioritise them accordingly as we manage and prioritise various products and projects it will allow the team of development and researchers to experiment and adapt various approach based on feedback that they receive regularly.

Benefits of Scrum

Now that we have seen the applications of scrum let's move on to discussing the benefits of the scrum framework, If the scrum framework is implemented in the right way it can offer numerous benefits for the organizations as well as the teams who are involved in the project management or product development let's see some key advantages of using the scrum framework:

1. Faster Development

- As we know that faster development is a crucial aspect of any project management scrum is used in order to achieve rapid development cycles there are several ways in which scrum helps us to achieve faster development some of which are iterative and incremental development,

- regular release cycles, prioritizing the features and quick adaptation to various changes made.
- Apart from that scrum also provides us with the facility to receive feedbacks through various ceremonies such as sprint review these rapid feedbacks allows the development team in order to identify and solve the issue faster.

2. Transparency in all Stages

In the scrum framework the transparency is a fundamental principle which enables clear communication at all the stages of a project being developed this transparency helps everyone such as the teams stakeholders as well as the organizations to understand the challenges progress as well as the outcome of their work in a scrum framework transparency is handled using various techniques such as product backlog sprint backlog sprint review daily standup meetings etc.

3. Time-efficient

Scrum framework is also known for its focus on the time efficiency because it uses the iterative development it allows for short cycles known as sprints which allows continuous improvement the scrum framework allows us to use several practices or principles which contribute to make sure that the projects we are making are more time efficient scrum handles time management through.

- Incremental development
- Clear roles and responsibilities
- Daily standup meetings
- Prioritisation and regular releases

Apart from that it also uses a concept of time boxed meetings in which various ceremonies such as the sprint planning, daily standup sprint review are time boxed meaning that they have a certain limit of time to finish these meetings the time boxing method allows us to do task in less time which ultimately leads to efficiency of time.

4. Budget-friendly Planning

The budgeting is an important role in the project management In order for us to maximise the revenue of the company we have to make sure that whatever product

or project we are building is budget friendly the scrum framework allows us to manage products within our budget because it follows various principles that helps us to prioritise the important factors of the project and it also allows us to reduce the wastage through following the incremental approach while dealing with new features.

Limitations of Scrum

As any other framework used in the product development the scrum also has some limitations in it as we have discussed applications and the advantages of using the scrum framework we also have to take into consideration of the limitations of this framework here are some common limitations of the scrum framework:

1. Not Suitable for All Projects

- The scrum framework will be best suited for projects which have challenging requirements but a high uncertainty if we have a project that is well defined and have stable requirements which will mean that once the project is ongoing, there is not much changes to be made in the requirements, Then it is advisable to use some other project management framework.
- As the scrum framework is more suited when the project has high uncertainty and the requirements are constantly changing so while dealing with a project that requires a strict and detailed plan then we can say that the scrum framework is not very well suited.

2. Requires Skilled & Experienced Team

- The scrum framework is only successfully implemented if the skill and experience of the people working on the project are high, the experience of the scrum team the product owner, scrum master and the development team makes a very big impact on the implementation of the scrum model.
- If we have any experience team and we implement the scrum model then the team may have struggle to fully understand the scrum principles and apply it which will lead to inefficiencies in the product development.

3. Limited Predictability

- The scrum framework or model has an adaptive nature which focuses on the changing requirements which makes it less predictable which can be limitation for some organisations because some organisations may

require strict predictability of the project timeline as well as the outcome for example some industries have regulatory compliance in which the use of scrum framework Is not possible.

4. Lack of Detailed Documentation

- The scrum model does not have a lot of documentation Which makes the development more time efficient as there is less time required for the documentation work but having less documentation can be challenging because in some industries it is required to have large amount of documentation such as for some pharmacies it is required to have extensive documentation in order to manage regulatory compliance so it can be a challenge to use the scrum model in such cases because this model does not provide large amount of documentation of the project.

5. Risk of Team Burnout

- As the scrum allows for continuous improvement using the iterative approach and allows for adaptability it can also create some pressure on scrum teams as we know it uses sprints cycles to meet the demands and deliver the request faster which may lead to burnout if the teams are not sufficient or if there is some lack of work life balance in the team, which can ultimately become disadvantage in the long run.

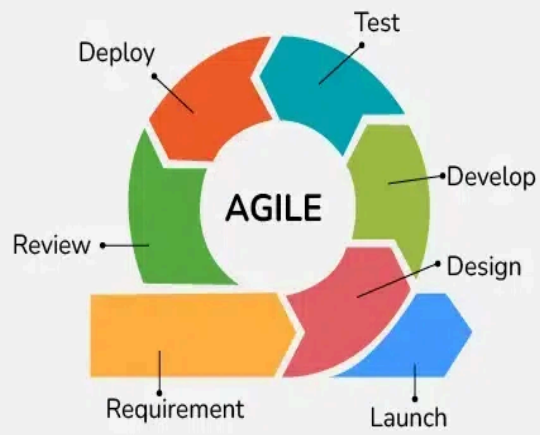
Agile Vs Scrum

Here is a table that represents the difference between agile and scum:

Parameters	Agile	Scrum
Methodology	Agile is a set of principles that's iterative and incremental.	Scrum is an implementation of the agile methodology.

Projects	Suited for projects involving a small team of experts.	They are used in projects where the requirements are constantly changing.
Leadership	The project head takes care of all tasks is vital to the project.	There's no leader, the scrum master, and the team addresses the issues. It involves cross-functional, self-organizing teams.
Flexibility	In agile, changes cannot be handled frequently.	It enables teams to react to changes quickly.
Delivery	The methodology requires frequent delivery to the end user.	With sprints, builds are delivered to clients for feedback.
Collaboration	Face-to-face interaction takes place between cross-functional teams.	Daily stand-up meetings help with collaboration.

Agile



VS

Scrum

